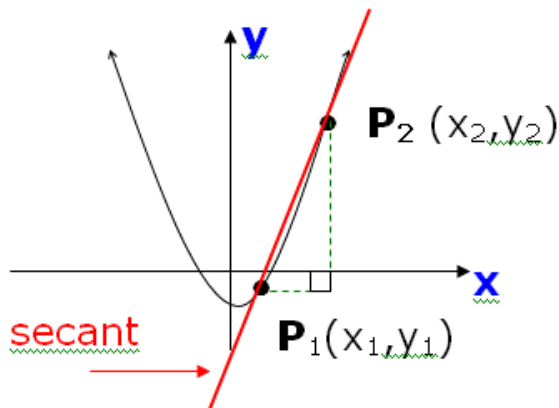


1.5: Slopes of Secants and Average Rate of Change

A rate of change:

- is a measure of how quickly one quantity (dependant variable) changes with respect to another quantity (independant variable)
- can be determined by calculating the slope of the secant passing through two points



Recall:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Example 1: The path of a basketball is modelled by the equation $h(t) = -4.9t^2 + 15t + 1.9$ where h represents height in metres and t represents time in seconds.

a) Determine the average rate of change of the height of the ball over the interval $0 \leq t \leq 2$.

b) Determine the average rate of change of the height of the ball over the interval $[3, 4]$.

Recall interval notation (see text page 8):

$[3, 4]$ ----- $3 \leq x \leq 4$ **and** $(3, 4)$ ----- $3 < x < 4$

c) Describe what the average rate of change means in each of the above situations.

Example 2:

A cube of ice is melting. Determine the average rate of change of melting if one side of the cube goes from 10 cm to 5 cm in 20 minutes.